MILITARY
SPECIFICATION

MIL-H-12240A(SigC)
18 August 1958
Superseding MIL-H-12240 (Si
1 October 1952

HYDROMETER, SYRINGE, BATTERY TS-765( )/U

1. SCOPE

1.1 Scope.- This specification covers a syringe type hydrometer, designated as Hydrometer, Syringe, Battery TS-765( )/U for determining the specific gravity, at 80/80°F, of lead-acid storage-battery electrolyte at temperatures from minus 65° to plus 165°F. (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-00591	Boxes, fiberboard, wood cleated
PPP-B-566	Boxes, folding paperboard
PPP-B-585	Boxes, wood, wirebound
PPP-B-601	Boxes, wood, cleated plywood
PPP-B-621	Boxes, wood, nailed and lock-corner
PPP-B-636	Boxes; fiber
PPP-T-76	Tape, pressure-sensitive adhesive, paper, water resistant

MILITARY

MIL-P-116	Preservation, methods of
MIL-M-13231	Marking of electronic items

STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and tables for inspection by attributes
MIL-STD-129	Marking for shipment and storage

DRAWINGS

SIGNAL CORPS

SC-C-33073	Form to be used for Pilot Pack Process Sheet

FSC 6630

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2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N. W., Washington 6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification

(Application for copies should be addressed to the Official Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Description.- Hydrometer Syringe, Battery TS-765( )/U is used for determining the specific gravity, at 80/80°F, of lead-acid storage-battery electrolyte at temperatures from minus 65°F to plus 165°F.

3.2 Parts, materials, and processes; general.-

3.2.1 Selection.- Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements, as specified herein.

3.2.1.1 Material.- Materials to be used in construction of the hydrometer shall be sound, of uniform quality and condition, free of defects, and shall conform in composition, heat treatment, and suitability to the standard practices of those manufacturers producing instruments of the type called for in this specification. Where a definite material is not specified for a particular application, the contractor shall select a material that will be entirely suitable for the purpose intended.

3.2.1.2 Metal components.- Metal parts, if used, shall be properly protected from acid fumes and electrolyte.

3.2.1.3 Rubber components.- Rubber parts shall be acid-resistant, show no discoloring and shall remain sufficiently pliable to function satisfactorily at low temperature when tested in accordance with 4.3.8. When tested at -65°F in accordance with 4.3.8, the bulb shall draw at least 35 cc of the electrolyte into the float chamber within 90 seconds. The rubber parts shall cause no sedimentation when tested in accordance with 4.3.9.

3.2.1.4 Glass components.- Glass components shall be of high grade glass, transparent, and with the exception of the top of the float, thoroughly annealed. The glass shall show no ring strain when viewed under a polariscope, and if any longitudinal strain is present, it must be faint and highly diffused. It shall be free from blisters, lumps, or other defects which will affect operation and life of the instrument. All glass components shall be capable of withstanding the thermal shock tests of 4.3.10.

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3.2.1.5 Bonding materials.- If adhesives are used for bonding any parts, they shall be resistant to sulfuric acid with specific gravity up to 1.400 (at 80°F) and capable of retaining adhesive qualities without weakening or becoming brittle over a temperature range of -65° to 165°F with up to 95 percent relative humidity.

3.3 Interchangeability.- The equipment shall be designed so that all corresponding parts of a given manufacturer's part number will be dimensionally and functionally interchangeable with each other. (See 4.3.4)

3.4 General design.- The detailed mechanical design of this equipment shall be accomplished by the manufacturer subject to the requirements of this specification; these requirements being detailed only to the extent considered necessary to insure the performance, and permanence in use of the desired mechanical characteristics. The design, layout, and assembly of the hydrometer and its component parts shall be such that, in the replacement of interchangeable parts, no adjustments are required that cannot be performed by a serviceman using standard tools customarily available.

3.5 Convenience in operation and maintenance.- The design shall be such as to provide maximum convenience to personnel in operating the hydrometer and in interchanging components between hydrometers.

3.6 Marking.-

3.6.1 General.- Marking shall conform to Specification MIL-M-13231.

3.7 Component parts.- The hydrometer shall consist of a barrel, thermometer, temperature-specific gravity correction scale, three round-bottom floats, a bulb for drawing in electrolyte, a plug to prevent the float from entering the bulb, an overflow tube to prevent overfilling the barrel with electrolyte, and a base or end closure with incorporated draw-in tube.

3.8 Barrel.- Barrel shall be made of glass, plastic or other suitable material of size, shape and construction to enable proper functioning and reading of the instrument, and to house and hold the float, the thermometer, and scales. It must protect the thermometer and scales from breakage and deterioration caused by moisture and acid. Length of float chamber shall provide for at least 1 inch free movement of the float in an axial direction.

3.8.1 Float chamber.- Inside diameter and shape of float chamber shall be such as to prevent air bubbles in the electrolyte from interfering with accurate reading of the float. Construction shall be such that when the instrument is completely assembled the float shall not come in contact with the barrel. Cleaning of the barrel, and its assembly to other parts of the instrument, shall be easily accomplished. Ends of the barrel shall be formed with beads or rolled edges to receive and hold securely the bulb and lower end closure of the instrument. Ample provision shall be made to insure proper flotation of float in solution being tested. The design must be such that a float scale reading cannot be obtained when the float is resting on the bottom of the float chamber. The amount of electrolyte necessary to obtain a reading shall not exceed 40 cc.

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3 8.2 Protective arrangements.- Means shall be provided to prevent float from entering hydrometer bulb and to prevent blockage of electrolyte being ejected. The top barrel plug shall be constructed with a rubber overflow tube. The overflow tube must serve to prevent the drawing up of electrolyte into the float chamber in such an amount as would cause the top of the float to contact the rubber plug in the top of the barrel Bulb and base must be designed to afford maximum protection to the barrel. Provision shall be made to protect the floats while instrument is being transported and used under any condition likely to be encountered in military service.

3 9 Floats.- Floats shall be of glass (3.2 1.4). Bulb and stem of floats shall each be of uniform diameter and circular in cross section. Lower portion of float body shall have a constriction to aid in the securing of the ballast. There shall be no deviation from axial alignment of the bulb ans stem apparent to the eye. Floats must float in a vertical position in an electrolyte of the highest specific gravity for which they are intended. Float accuracy over the entire scale shall be plus or minus 5 points (.005) specific gravity.

3.9.1 Flotation.- The free float shall normally assume a vertical position and shall not deviate from the vertical by more than 1/8 inch at the upper tip of the float.

3.9.2 Float scales.- Scale on floats shall be graduated for reading of specific gravity at 80/80°F, and shall be not less than 1-3/4 inches long, marked in increments of ten points (0.010) specific gravity and numbered each twenty points. Numbers shall appear in not less than two places on the scale, neatly printed in black on good grade white paper or plastic, and shall be easily readable under ordinary light conditions, with the instrument in a vertical position. With the float in a vertical position the lines shall be horizontally marked and shall bisect the numbers denoting their specific gravity values.

3.9.2.1 Float designations.- There shall be three floats, designed as Tropical, Temperate, and Arctic, and having the following ranges:

Float Designation	Sp. Gr. (80/80°F)
Tropical	1.040 to 1.280
Temperate	1.120 to 1.360
Arctic	1.180 to 1.400

3.9.3 Ballast.- Ballast shall be shot, sealed in place at the constriction in the float body and secured in such a manner as to prevent its loosening in the temperature range of -65° to +165°F. Ballast is considered loose when it has enough movement to cause the hydrometer to read inaccurately.

3.9.4 Replacement data.- Replacement data shall be marked on each float scale, showing the manufacturer's name, the type of float, (Tropical, Temperate, or Arctic) the military designation, and other information required to facilitate replacement.

3.10 Bulb.- Bulb shall be of such size, capacity, material and design as to insure that sufficient electrolyte shall be drawn into the barrel, under all climatic conditions, at any temperature within the range of the instrument, to insure accurate reading of hydrometer float at any point within the scale range. A groove shall be formed in bulb to fit over end of barrel. The bulb shall be designed to afford maximum protection from dropping or impact to the barrel. An eyelet for hanging the instrument up must be molded in the bulb.

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3.11 Plug.- A rubber plug shall be provided (3.8.2) and arranged in bulb end of barrel to protect the float and prevent its entrance into bulb of instrument.

3.12 Base and Draw-In Tube.- A rubber base with incorporated draw-in tube shall be securely fitted to the lower end of the barrel. The base shall be designed to afford maximum protection from dropping and impact to the barrel. The base shall also be designed to prevent the instrument from rolling when placed on a smooth surface. The draw-in tube shall be flexible rubber integral with the base, 4-3/4 $\pm\frac{1}{4}$ inches long (with outside diameter of $0.313\ ^{+.010}_{-0}$ inch and inside diameter of $0.125\ ^{+.010}_{-0}$ inch) designed to enable proper checking of any standard free-electrolyte type lead-acid storage battery. Temperatures within the range of the instrument shall have no detrimental effect on the draw-in tube or cause it to become hard. The bumpers projecting from the rubber base shall contain holes or a molded eyelet for hanging the instrument in an inverted position. The hydrometer shall not (a) permit drippage in excess of 0.5 cc during the first minute, nor more than 1.0 cc total during the test per 4.3.11.1, and (b) permit drippage in excess of 0.25 cc during the first minute, nor more than 0.5 cc total during the test per 4.3.11.2.

3.13 Thermometer.- The thermometer shall be calibrated at total immersion and shall accurately indicate the temperature of electrolyte being tested within plus or minus 7°F when tested in accordance with 4.3.7. It shall be securely fixed in position and shall be protected from breakage. The scale length, -65° to +165°F, shall be a minimum of 2-7/8 inches.

3.13.1 Thermometer and correction scales.- The thermometer and specific gravity correction scales shall be printed on good quality paper or plastic arranged on opposite sides of the thermometer. Markings shall be plain and as large as practicable. All numbers shall be horizontally bisected by the lines coinciding with their values. Brief but definite instructions for the correct computation of specific gravities at 80°F shall be printed to the right of the specific gravity correction scale.

3.13.2 Thermometer scale markings.- Markings shall be adjacent to and at the left of the thermometer stem (when facing instrument), in increments of 5°F and numbered each ten degrees from -65° to +165°F. Figures on the thermometer scale shall be black at temperatures above 0°F and red at temperatures below 0°F.

3.13.3 Correction-scale markings.- Markings shall be adjacent to and at the right of the thermometer stem (when facing instrument). Divisions on the scale shall represent not more than two points (0.002) specific gravity, accurate to one division and numbered in groups of not more than two divisions above and below the division corresponding to a temperature of 80°F. Correction shall be based on assumption that specific gravity changes one point (0.001) for each 2.5°F. The scale shall indicate which readings are to be added to or subtracted from the observed specific gravity to obtain the correct value at 80°F. Specific gravity correction figures to be added shall be black; those to be subtracted shall be red. Thus a float reading of 1.240 taken at 120°F should be corrected by +16 (as shown in black) to give a gravity reading of 1.256, and a float reading of 1.240 at -35°F should be corrected by -46 (as shown in red) to give a gravity reading of 1.194.

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3.14 Container.- A substantial, durable container shall be furnished to house the assembled instrument and floats. Its design shall be such as to preclude damage to the instrument while being stored or transported. Instructions, together with the following chart showing freezing points of fully or partly charge batteries, shall be packed with each hydrometer.

CHART

Condition of Charge	Tropical Float 1.040 - 1.280		Temperate Float 1.120 - 1.360		Arctic Float 1.180 - 1.400	
	Sp. Gr. 80/80°F	FP* °F	Sp. Gr. 80/80°F	FP* °F	Sp. Gr. 80/80°F	FP* °F
Full	1.220	-25	1.280	-92	1.330	-85
3/4	1.180	- 4	1.246	-58	1.295	-95
1/2	1.154	+ 6	1.214	-20	1.265	-80
1/4	1.222	+14	1.184	- 5	1.232	-40
Discharged	1.090	+19	1.150	+ 7	1.200	-12

* FP denotes freezing point of the acid solution.

3.15 Dimensions.- The overall length of the hydrometer, exclusive of the draw-in tube (3.12) shall not exceed 12 inches. The maximum dimensions of the base shall be 3 1/8 inches long x 3 1/8 inches wide. The diameter of the bulb shall not exceed 3 1/8 inches.

3.16 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.1(e)(1)).

3.17 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.1).

3.18 Operation.- The hydrometer shall be of such size and design that when the rubber bulb is pressed between the thumb and two fingers until the inside walls touch and is then released, sufficient electrolyteshall be drawn into the barrel to permit a accurate reading of the hydrometer (or float) even thought the electrotyte has a specific gravity equal to the lowest value on the hydrometer scale. The readings (specific gravity and temperature) shall meet the accuracies specified in 3.9 and 3.13.

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3.19 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with applicable portions of the following paragraphs herein:

Paragraph 3.2 Parts, materials, and processes; general

Paragraph 3.4 General design
Paragraph 3.5 Convenience in operation and maintenance
Paragraph 3.6 Marking
Paragraph 3.7 Component parts
Paragraph 3.8 Barrel
Paragraph 3.8.1 Float chamber
Paragraph 3.8.2 Protective arrangements
Paragraph 3.9.1 Flotation
Paragraph 3.9.2 Float scales

Paragraph 3.9.2.1 Float designations
Paragraph 3.9.3 Ballast
Paragraph 3.9.4 Replacement data
Paragraph 3.10 Bulb
Paragraph 3.11 Plug
Paragraph 3.13 Thermometer
Paragraph 3.13.1 Thermometer and correction scales
Paragraph 3.13.2 Thermometer scale markings
Paragraph 3.13.3 Correction scale markings
Paragraph 3.14 Container
Paragraph 3.15 Dimensions
Paragraph 3.18 Operation

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.4.)

(1) Procurement inspection of items before preparation for delivery. (See 4.3.)

(2) Procurement inspection of preparation for delivery. (See 4.4 and section 5.)

4.2 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and the group A, and group B, inspection specified in table II and III, respectively, except for the following destructive tests:

Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements.

Table I.- Preproduction inspection

Inspection	Reqt Par.	Test Par.
Acid resistance	3.2.1.3	4.3.9
Low Temperature	3.2.1.4	4.3.8
Thermometer	3.13	4.3.7

4.3 Procurement inspection of items before preparation for delivery.- The minimum inspection performed by the contractor shall be as specified by (a) through (c) below-- to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.5) Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A and group B, inspection as specified in 4.3.1 through 4.3.2.4, and shall be performed by the contractor.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

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(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group a inspection shall be performed in any order which is satisfactory to the Government, except the operational inspection (4.3.6) shall be last.

Table II.- Group A Inspection

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
Visual & Mechanical Hydrometer, Syringe, Battery TS-765( )/U Container	3.19 3.14	4.3.4 4.3.4	1.0%	4.0%
Operational	3.8 3.11 3.12 3.19	4.3.6 4.3.11	1.0%	

4.3.2 Group B inspection.- This inspection (including sampling) shall conform to table III and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.3 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table III, shall be as follows:

Group B plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B-1	4.0%	L-8	L-6
B-2	6.5%	L-7	L-5
B-3	10.0%	L-6	L-4
B-4	15.0%	L-5	L-3

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

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4.3.2.3 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.3.2.4 Reinspection of conforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.3 Disposition of nonconforming product.- When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

Table III.- Group B inspection

Inspection	Rqt. Par.	Test Par.	Group B (see 4.3.2.1)
			B-2
Thermal Shock	3.2.1.4	4.3.10	6.5% AQL
			L-7 N & T
			L-5 R
Float	3.9	4.3.5	B-2
Ballast	3.9.3	4.3.7	
Interchangeability	3.3	4.3.12	B-2

4.3.4 Visual and mechanical inspection.- The equipment shall be inspected for material, workmanship, mechanical fit, glass strain, printing, identification marking, and miscellaneous defects.

4.3.5 Float tests.- Before assembly of hydrometers, the floats shall be floated in an open vessel to facilitate comparison with a certified precision standard hydrometer. Each float must be tested individually by floating it in the test solutions simultaneously with the standard hydrometer. The floats shall comply with all provisions of 3.9 when tested in solutions adjusted to specific gravities corresponding approximately to the extremes and middle of each float's scale.

4.3.6 Operational test.- A solution shall be drawn into the hydrometer from an open vessel containing a precision standard hydrometer and thermometer. After several syringings to obtain temperature equilibrium within the hydrometer, the readings of the float and thermometer under test shall be taken. (See 3.18)

4.3.7 Thermometer test.- The thermometer shall be tested at temperatures of -65°F, -32°F, 0°F, and 32°, 80°, 120° and 165°F by comparison with a certified thermometer used simultaneously in the same solution, and shall meet the accuracy requirements of paragraph 3.13. In order to insure temperature equilibrium within the instrument before a reading of the thermometer is taken, the float chamber must be flushed at least three times with solution from a constant temperature bath containing the standard thermometer.

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During this test the hydrometer shall be immersed in the constant temperature bath to a depth beyond the top of the thermometer scale at -65°F and +165°F the ballast shall be inspected and shall meet the requirements of 3.9.3.

4.3.8 Low Temperature.- Subject the hydrometer to -65°F for 290 hours. Then to determine the flexibility of the bulb, draw in at least 35 cc of electrolyte, (1330 specific gravity 80°/80°F) which has been cooled to -65°F. The electrolyte shall be drawn into the float chamber and conform to the requirements of 3.2.1.3.

4.3.9 Acid resistance test.- The hydrometer shall be disassembled, and all parts, except the float, shall be immersed in sulphuric acid of 1.400 specific gravity at 80°F for 96 hours. At the end of that period examination shall be made to determine compliance with the requirements of paragraphs 3.2.1.3 and 3.2.1.5.

4.3.10 Thermal shock test.- With hydrometer stabilized at 80°F, draw in at least 35 cc of electrolyte that has been cooled to -65°F and expel. Repeat this procedure three times to determine compliance with 3.2.1.4.

4.3.11 Drippage Tests.-

4.3.11.1 Loaded Barrel Drippage Test.- Submerge the tip of the draw-in tube approximately one-half inch in the electrolyte reservior, depress the bulb sufficiently to draw at least 35 cc of electrolyte into the barrel. Release the bulb and allow it to resume its normal shape. Hold the hydrometer in an upright position with the tip submersed for 100seconds ±5 seconds. Raise slowly from the reservoir, suspend above reservoir, and observe and meter the drippage at the end of (a) 1 minute of suspension, and (b) 10 minutes of suspension. Expel the electrolyte from the hydrometer preparatory to performance of the test 4.3.11.2. (See 3.12)

4.3.11.2 Empty Barrel Drippage Test.- After filing the hydrometer as described in 4.3.11.1, hold the hydrometer vertically and expel the electrolyte by deflating the bulb as completely as practicable by hand, then release the bulb. Repeat twice at approximately 3 second intervals (totaling 3 bulb deflating actions.) Upon completion of the third bulb release, observe and meter electrolyte drippage (a) at 1 minute after the third release, and (b) at 10 minutes after the third release. (See 3.12)

4.12 Inspection for interchangeability.- The componants listed below shall be gaged or measured (see table III) to determine compliance with the physical interchangeability requirement of 3.3. When a listed componant is not within specified or design limits, it shall be considered a major defect.

(a) Syringe Rubber Bulb
(b) Barrel
(c) Floats

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4.4 Procurement inspection of preparation for delivery.- The minimum inspection performed by the contractor shall be as specified. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.5) Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.4.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table IV.

TABLE IV

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Any item of marking information other than 5 thru 7 omitted, incorrect, or illegible.
3. Incorrect packaging method applied.	

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TABLE IV (CONTD)

MAJOR	MINOR
4. Cushioning inadequate for the physical and mechanical protection of the item.	
5. Stock number omitted, incorrect, or illegible.	
6. Nomenclature omitted, incorrect, or illegible.	
7. Marking of quantity of items in package omitted, incorrect or illegible.	
8. Difference stock numbered items in the same unit package.	

4.4.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table V.

TABLE V

PACKING, AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Gross weight in excess of specified amount.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Box closure not as specified.	4. Any item of required marking information other than 9 th 15 listed under major defect omitted, incorrect, or illegible.
5. Type, grade, class, and style of the shipping container not as specified.	
6. Strapping omitted.	
7. Strapping inadequate or incorrectly applied.	
8. Shipping documents or packing list omitted.	
9. Stock number omitted, incorrect, or illegible.	
10. Nomenclature omitted, incorrect, or illegible.	
11. Marking of quantity of items in pack omitted, incorrect, or illegible.	

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TABLE V (CONTD)

	MAJOR	MINOR
12.	Destination marking omitted, incorrect, or illegible.	
13.	Service designation (color marking) omitted.	
14.	Specified special marking and labeling such as NAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
15.	Overseas code marking omitted, incorrect, or illegible.	

4.4.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.4.2.2 Procedure in case of failure.- If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.4.2.3 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Rough handling test (preparation for delivery).- When rough handling test is required by the contract (see 6.1 (e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Hydrometer, Syringe, Battery TS-765( )/U shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Hydrometer, Syringe, Battery TS-765( )/U.-

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5.1.1.1.1 Unit packaging.- Each hydrometer in its container (see 3.14) shall be unit packaged individually Method III as follows: Each container housing the assembled syringe and floats shall be placed within a close-fitting paperboard box, style II, type A, class (a) conforming to Specification PPP-B-566. Box closure shall be in accordance with the requirements of the appendix of the referenced box specification.

5.1.1.2 Intermediate packaging.- A uniform quantity of unit packages specified in 5.1.1.1.1 shall be placed within a close-fitting regular slotted style class 2W6c fiber box conforming to Specification PPP-B-636. Box closure shall be effected by the applicati of pressure sensitive water-resistant tape conforming to Specification PPP-T-76, to all seams and joints of the box. The weight of box and contents shall not exceed 30 pounds.

5.1.2 Level C. A uniform quantity of Hydrometers, Syringe, Battery TS-765( )/U shall be packaged in accordance with commercial practice, and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Hydrometer, Syringe, Battery, TS-765( )/U shall be packed for shipment as follows: A uniform quantity of intermediate packages (see 5.1.1.2) shall be packed within an overseas type or class wirebound wood box a wood-cleated fiberboard box, a wood cleated plywood box, or a wood nailed and lock-corner box, conforming to Specification PPP-B-585, PPP-B-00591, PPP-B-601, or PPP-B-621, as applicable. Boxes shall be fabricated to fit the contents snugly. The weight of any one box with contents shall not exceed 200 pounds. Box closure shall be in conformance with the requirements of the appendix of the applicable box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Hydrometer, Syringe, Battery TS-765( )/U shall be packed for shipment as specified in 5.2.1 except boxes shall be domestic type or class.

5.2.3 Level C. A uniform quantity of Hydrometers, Syringe, Battery TS-765( )/U shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4, and Consolidated Freight Classification Rule 21, for rail shipment, National Motor Freight Classification No. A-3, and No. 14 for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A or level B.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 2 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.16.)

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.17)

(i) Quantity of tools and running spare parts required. (See 3.17)

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6.2 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PIQ-1W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.3 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.4 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspecti

6.4.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.5 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility no any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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